Changelog

1.2.1

Fixes from a project-wide review triggered by a real degraded-state incident (old listener + new server = every payload empty while status read OK).

- **CRITICAL: switched to ThreadingHTTPServer.** The v1.2.0 listener used a single-threaded HTTPServer, so while `/exec_sync` held the one HTTP thread waiting on a job (up to ~110s), `/ping` and `/state` could not answer - silently defeating the "separate always-responsive channel" design. Verified fixed: ping now answers instantly during a blocked sync call.
- **Version/protocol handshake.** The listener now reports `listener_version` + `protocol` in every response; the server checks it and returns a loud PROTOCOL MISMATCH message with exact fix steps when the add-in is stale - instead of silently rendering empty payloads as STATUS: OK. This makes the incident that prompted this release impossible to miss.
- **Fail-loud capture health.** Envelopes include `stdout_bytes`; the server marks ok-but-silent results with "(no output captured)".
- **Job-store pruning.** Finished jobs are capped at the 50 most recent so long sessions don't grow memory unboundedly.
- **Docs:** INSTALL now warns that Fusion loads the add-in from its AddIns folder, not from wherever the project files live - updating one does not update the other.

1.2.0

Async execution and document safety — driven by a second round of real-use feedback. The core fix: a 4-minute timeout that told you nothing about what actually happened in Fusion. Execution is now decoupled from the HTTP response.

Added — async job model (the big one)

- **Listener is now async.** `POST /exec` enqueues a job and returns a `job_id` immediately; the operation runs on the main thread and is polled via `GET /job/<id>`, which returns a structured envelope (`status`, `stdout`, `traceback`, `elapsed_ms`) and **partial stdout while still running** (incremental progress).
- **`run_async(code)` + `job_status(job_id)`** tools for long operations that shouldn't block. Plus a `progress(msg)` helper to mark stages visible mid-run.

- **Ping/state on a separate channel.** `GET /ping` and `GET /state` are served from cache without the main thread, so they STAY RESPONSIVE during (or after) a hung operation. A hung op and a dead listener are now distinguishable.
- `get_state()` — cheap, guaranteed-fast document identity + counts (no geometry work). The safe first call for crash recovery.
- **Structured result envelope, always.** Every execution returns status/stdout/traceback/elapsed; no more bare "no result received". The sync path returns partial output and a job id if the op outlives the wait.
- **Watchdog** flags jobs that exceed their budget as `timeout` (clear signal rather than a dead channel).

Added — document safety

- **Document identity surfaced everywhere.** `get_state` and `get_model_state` lead with the active document **name + lineage id**, preventing wrong-document overwrites.
- `save_document(name, create_new=False)` now reports **which lineage it versioned**, and `create_new=True` saves a brand-new lineage so a new part never buries an unrelated design's history.
- `list_versions(design_id="")` — read-only enumeration of a lineage's saved versions, so a bad save is recoverable and the user can be pointed to the exact version to promote.
- `open_document` reports the new active lineage.

Added — documentation

- Sketch-plane / axis mapping (XY/XZ/YZ + offset planes, incl. the XZ→world-−Z inversion) documented in `run_fusion_code`'s docstring.

Notes

- `run_fusion_code` uses the bounded sync path (~100s) then hands off to polling; no behavior change for short ops.
- Volume-delta no-op detection and the `mm()` unit helper shipped in 1.1.0 and already cover two earlier feedback items.

1.1.0

Observability, diagnostics, and stability. Injected unit/observability helpers (`mm()`, `assert_scale()`, `report_volume_delta()`, etc.); auto volume-delta reporting with no-op warnings on cut/shell tools; `cross_section()`, `list_faces()`, `list_edges()`, `heartbeat()`; listener-level refusal of known host-crashing calls; richer `get_model_state()`.

1.0.0

First public release. Fusion 360 add-in listener + MCP server with ~30 tools (primitives, parameters, faces, assemblies/joints, measurement, visual feedback, export, persistence), preflight launcher, and full docs.